

Brainstorming & Idea Prioritization

Filled project document for GitHub and SkillWallet submission

Date	28 June 2026	Team ID	To be updated in SkillWallet
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction	Team Members	Devireddy Prasanna, Chokka Abitha, Navya Borugadda, Bodi Varshitha, Mulakala Harika Venkata Sridevi

1. Team Gathering and Selected Problem Statement

Flood-prone communities and disaster management teams often receive warnings too late because conventional forecasting workflows do not always convert weather observations into fast, usable risk classifications. The selected problem is to create an accessible machine learning system that predicts flood likelihood from rainfall and weather parameters.

2. Brainstormed Ideas

Idea	Description	Expected Value
ML-based risk classifier	Train supervised models using historical rainfall, visibility, and seasonal weather data.	Fast flood/no-flood prediction.
Flask dashboard	Provide a simple web form for meteorologists and officials to enter regional readings.	Easy access without notebook knowledge.
Prediction history	Store recent predictions for review and audit.	Improves monitoring and accountability.
Cloud deployment	Deploy the application on IBM Cloud for public or controlled access.	Scalable availability for authorities.
Model comparison	Compare Decision Tree, Random Forest, KNN, and XGBoost.	Select the most reliable algorithm.

3. Prioritized Solution

The team selected a supervised machine learning flood prediction web application because it directly addresses early warning, resource planning, and model validation scenarios. XGBoost is expected to be the preferred model if it maintains the highest validation accuracy.